DeepShop: A Benchmark for Deep Research Shopping Agents Supplementary Material

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Appendix

A Details of Dataset Analysis

A.1 Constraint counts across domains

We analyse the 50 original seed records used in constraint counting and the full 600-query candidate pool, including those seed records. Table 1 reports the mean total count of product attributes, search filters, and sorting preferences within each domain.

Macro avg. and Domain SD are the unweighted mean and population standard deviation of the five domain means. The candidate pool has a higher mean constraint count and less variation between domain means than the counted seed records.

Table 1. Overall task complexity distribution across domains.

Data	Electronics	Fashion	Sports	Books	Home	Macro avg.	Domain SD
Seeds (50)	3.43	3.60	3.29	4.25	3.50	3.61	0.33
Pool (600)	6.22	6.30	6.32	6.26	6.11	6.24	0.08

A.2 Manual task validation

GPT-4o generates candidate queries. All 150 retained tasks are manually executed on live Amazon to check coherent intent, satisfiable constraints, and task completion at construction time. Unnatural queries are rewritten, while candidates failing these checks are revised or discarded. These checks establish task feasibility at construction time.

Table 2. Constraint-type coverage of the 150 tasks.

Dimension	Types	Most frequent type
Product attributes	18	Brand/model (14%)
Search filters	17	Rating ≥ 4.5 stars (19%)
Sorting preferences	14	Highest-rated (22%)

A.3 Lexical and semantic diversity

Following WebVoyager [3], we compute pairwise semantic similarity among the 150 retained queries using `all-mpnet-base-v2`; 97.64% of query pairs have similarity below 0.6. Lexical overlap is measured by word-level Jaccard similarity, the size of the intersection of two query word sets divided by the size of their union. Mean pairwise overlap is 0.17 for the retained queries and 0.13 for the 50 real-user seed queries. These measures quantify semantic and lexical redundancy.

A.4 Constraint-type coverage

We classify the constraints of the 150 retained tasks within the three evaluation dimensions. Table 2 summarizes the number of types and the most frequent type in each dimension. Percentages are calculated over constraints within the corresponding dimension; the most frequent type accounts for at most 22%.

B Details of Evaluation

B.1 GPT-4o for Evaluation

This section details the GPT-4o prompt used for fine-grained evaluation. As shown in Figure 1, we adapt the prompt from prior work [3] to assess whether the web agent satisfies the requirements related to product attributes, search filters, and sorting preferences. For each trajectory, GPT-4o receives a dimension-specific subquery, up to 15 screenshots captured during the run, and the agent's final answer.

For each applicable dimension, the evaluator returns a binary decision ("Success" or "Not Success"). A missing subquery is excluded from that dimension's aggregate score and treated as satisfied for holistic evaluation. Overall task success is the logical conjunction of the applicable dimension-level decisions, requiring all stated requirements to be satisfied jointly.

B.2 Human evaluation

We report the original human comparison of GPT-4o judgments on 50 WebVoyager trajectories. Following previous work [3,6], we hire human annotators with at least an undergraduate degree, and each sampled trajectory is independently evaluated by a minimum of two annotators. In cases of disagreement, a third annotator is consulted, and the final judgment is determined by majority vote. The evaluation is conducted

Table 3. Paired comparisons of original-run task success.

Comparison	Difference (pp)	95% CI (pp)	p
Browser Use – SeeAct	21.33	[13.27, 29.39]	5.62×10^{-7}
Browser Use – Simple RAG	24.67	[16.53, 32.80]	1.49×10^{-8}
Browser Use – Agent-E	25.33	[17.81, 32.85]	5.04×10^{-10}
SeeAct – Simple RAG	3.33	[−2.40, 9.07]	0.253
SeeAct – Agent-E	4.00	[−1.25, 9.25]	0.134
Simple RAG – Agent-E	0.67	[−4.10, 5.43]	0.783

based on the full execution trace of the agent, including the task description, operation history, and screenshots, as shown in Figure 2. Each human annotator labels the success of each subgoal—product attributes, search filters, and sorting preferences—as well as the overall task completion. The average Cohen's κ among human annotators is 0.82. Human–GPT-4o agreement is 84%, 80%, 82%, and 86% for product attributes, search filters, sorting preferences, and overall task success, respectively.

Additionally, since deep research systems do not expose intermediate screenshots or action histories, human annotators access Amazon to verify whether the returned links satisfy the specified requirements. The original commercial-system evaluation used a 30-query subset, with the first recommended item evaluated when multiple products were returned. These runs supply the historical cases in Section E. The updated commercial results in the main table instead cover all 150 queries using the API versions specified in the main text.

B.3 Significance test between web agents

We recompute two-sided paired t-tests from saved binary task-success outcomes for Simple RAG, Agent-E, SeeAct, and Browser Use on the same 150 task IDs. Differences are oriented from the higher to the lower observed success rate. Table 3 reports unadjusted p-values and pointwise 95% paired-t confidence intervals ($df = 149$). These exploratory comparisons are restricted to the listed original runs; they are not tests of the commercial APIs. Grouping the same outcomes by category or difficulty reproduces the four original-baseline series in Figure 3 of the main paper and exactly recovers their corresponding overall scores.

C Details of Baselines

– **Simple RAG method**: We implement a simple retrieval-augmented generation (RAG) baseline by combining GPT-4o with Google Search. The user query is first submitted to Google Search, and the top-ranked webpage is selected. GPT-4o (version 2024-08-06) then generates a final response conditioned on the screenshot of the retrieved page. We use the Serper API[1] to programmatically access Google search results.

[1] https://serper.dev/

[System prompt]
As an evaluator, you will be presented with three primary components to assist you in your role:

1. Web Task Instruction: A clear and precise natural language directive that specifies an online shopping activity to be executed. The instruction may involve locating products that meet certain attribute requirements (e.g., color, size, brand), applying specific search filters (e.g., price range, customer ratings, availability), or fulfilling user-defined sorting preferences (e.g., lowest price, newest arrivals, best sellers). Tasks may also include verifying product details, comparing offers, or checking for shipping and return policies, depending on the scenario.

2. Result Screenshots: This is a visual representation of the screen showing the result or intermediate state of performing a web task. It serves as visual proof of the actions taken in response to the instruction.

3. Result Response: This is a textual response obtained after the execution of the web task. It serves as textual result in response to the instruction.

-- You DO NOT NEED to interact with web pages or perform actions such as conducting searches on websites.
-- You SHOULD NOT make assumptions based on information not presented in the screenshot when comparing it to the instructions.
-- Your primary responsibility is to conduct a thorough assessment of the web task instruction against the outcome depicted in the screenshot and in the response, evaluating whether the actions taken align with the given instructions.
-- NOTE that the instruction may involve more than one task, for example, locating the garage and summarizing the review. Failing to complete either task, such as not providing a summary, should be considered unsuccessful.
-- NOTE that the screenshot is authentic, but the response provided by LLM is generated at the end of web browsing, and there may be discrepancies between the text and the screenshots.
-- Note the difference: 1) Result response may contradict the screenshot, then the content of the screenshot prevails, 2) The content in the Result response is not mentioned on the screenshot, choose to believe the content.

You should elaborate on how you arrived at your final evaluation and then provide a definitive verdict on whether the task has been successfully accomplished, either as 'SUCCESS' or 'NOT SUCCESS'.

[User prompt]
TASK:
{subquery}

Result Response:
{answer}

15 screenshots at the end:
{screenshots}

Fig. 1. Prompts for GPT-4o evaluation

You will be presented with a web shopping task.
For each task, you will receive three subqueries, along with the web agent's action history and corresponding screenshots. Your goal is to evaluate the agent's performance across three specific dimensions: product attributes, search filters, and sorting preferences.
Please note: if a subquery is labeled as None, you do not need to assess that particular aspect.

Definitions of the three subqueries are as follows:
1. Product attributes, referring to concrete product characteristics users may specify to express detailed intent
2. Search filters, representing categorical or numerical constraints commonly used on e-commerce platforms
3. Sorting preferences, indicating desired result orderings, such as price or popularity.

Task:
{Query}

Product Attribute Requirement:
{Subquery1}

Search Filter Requirement:
{Subquery2}

Sorting Preference Requirement:
{Subquery3}

Agent Action History:
{Action}

Screenshots:
{Screenshots}

Fig. 2. Instructions for human evaluation.

- **Web agents**: The original experiment record identifies GPT-4o for WebVoyager, Browser Use, and SeeAct, and GPT-4o-mini for Agent-E. **Agent-E** [1] is an HTML-based agent that adopts a hierarchical planner-actor architecture, augmented with flexible DOM tree distillation and a denoising mechanism to improve decision accuracy.[2] **SeeAct** [7] exploits the multi-modal capabilities of large language models (LLMs), integrating visual perception with structured web-based interactions.[3] **WebVoyager** [3] also leverages multi-modal reasoning and introduces a set-of-mark prompting scheme that guides the agent to first generate intermediate thoughts before selecting final actions.[4] **Browser Use** [4] is an open-source web agent framework that combines visual understanding with HTML structure parsing to support robust web navigation and interaction.[5]
- **Deep research systems**: Since we cannot strictly constrain deep research systems to operate on specific shopping websites, the original commercial evaluations include explicit site constraints in the prompt to guide the search process (Figure 3). The original 30-query diagnostic runs used **Gemini Deep Research** [2] with Gemini 2.0 Flash[6] and the original **OpenAI Deep Research** [5] with o3.[7] These historical versions supply the commercial cases in Section E. They are distinct from the full-150-query evaluations using `deep-research-preview-04-2026` and `o3-deep-research` in the updated main table.

Prompt:
Your task is to help find relevant products on Amazon website (https://www.amazon.com/?language=en_US&currency=USD) based on the following question: {Query}

Please strictly follow the rules:
1. Do not contact the user for further information. Provide answers directly without asking any questions or seeking clarification.
2. Include Amazon product links for all recommended items.

Fig. 3. Prompts for deep research systems.

D Details of Implementation

D.1 Prompts for query diversity and complexity evolution

Details for the prompts used in Diversity$(\cdot)$, and Complexity$(\cdot)$ are provided. Figures 4, 5, and 6 display the prompts for query diversity, complexity evolution and detailed complexity evolution strategy, respectively.

[2] https://github.com/EmergenceAI/Agent-E
[3] https://github.com/OSU-NLP-Group/SeeAct
[4] https://github.com/MinorJerry/WebVoyager
[5] https://github.com/browser-use/browser-use
[6] https://blog.google/products/gemini/google-gemini-deep-research/
[7] https://openai.com/index/introducing-deep-research/

D.2 Implementation details

The original experiments evaluate Agent-E, SeeAct, WebVoyager, and Browser Use in live web environments. Agent-E, SeeAct, and Browser Use use Playwright, while WebVoyager uses Selenium. The original protocol specifies a maximum of 15 steps per task and a `gpt-4o-2024-08-06` evaluator at temperature 0. Model information is provided in Section C. The original implementation describes full-page screenshots for Agent-E and SeeAct and visible-viewport screenshots for WebVoyager and Browser Use.

```
I want you act as a Prompt Creator.
Your goal is to draw inspiration from the #Given Prompt# to create a brand new prompt.
This new prompt should be in the web shopping domain but tailored for different specific
products in the {Randomly selected product category} amazon product field.
The LENGTH and complexity of the #Created Prompt# should be similar to that of the
#Given Prompt#.
The #Created Prompt# must be reasonable and must be understood and responded by
humans.
'#Given Prompt#', '#Created Prompt#', 'given prompt' and 'created prompt' are not allowed to
appear in #Created Prompt#

#Given Prompt#:
{Original query}

#Created Prompt#:
```

Fig. 4. Prompts for query diversity evolution.

```
I want you act as a Prompt Rewriter for web shopping.
Your objective is to rewrite a given prompt into a more complex version to make those web
shopping agents a bit harder to handle.
But the rewritten prompt must be reasonable and must be understood and responded by
humans.
You should complicate the given prompt using the following method:
{Randomly selected evolution strategy}
You should try your best not to make the #Rewritten Prompt# become verbose, #Rewritten
Prompt# can only add 10 to 20 words into #The Given Prompt#.
'#The Given Prompt#', '#Rewritten Prompt#', 'given prompt' and 'rewritten prompt' are not
allowed to appear in #Rewritten Prompt#

#Given Prompt#:
{Original query}

#Created Prompt#:
```

Fig. 5. Prompts for query complexity evolution.

Product attribute evolution:
Enhance #The Given Prompt# by integrating detailed product attributes that detail user needs. Please specifies concrete values for one product attribute (e.g., brand, model, price range, color, size, weight, or unique features) based on your knowledge about this product, ensure that these exact details are incorporated into the query instead of using generic placeholder terms.

Search filter evolution:
Enhance #The Given Prompt# by integrating detailed product constraints that capture user needs. Please specifies concrete values for constraints—such as a minimum customer rating (e.g., above 4.0 or 4.5 stars), a minimum number of customer reviews (e.g., 100, 300, 500, or 1000), shipping options like free delivery, new arrival time frames (e.g., released in the last 30 or 90 days), return policies (e.g., free returns), or warranty information (e.g., includes a 1-year warranty) based on your knowledge about amazon website, ensure that these exact values are used in the query rather than generic terms.

Sorting preference evolution:
Enhance #The Given Prompt# by integrating a specific product filtering requirement for web shopping. Find the top product based on one of the following criteria: lowest price, highest user rating, newest arrival, or best seller ranking.

Fig. 6. Prompts for complexity evolution strategies.

E Details of Error Analysis

These examples come from the original web-agent runs and the earlier 30-query commercial-system evaluation. They illustrate failure mechanisms in those versions.

- **Web agents are limited by grounding ability.** As shown in Figure 7, agents struggle to accurately ground interface elements. For example, button 39 related to user rating was not properly segmented, preventing the agent from selecting a specific rating range. Buttons 31–37 and 41–44 were rendered too densely and overlapped, making interaction difficult. Additionally, the sorting button on the right was incorrectly split into two buttons 16 and 17, which may confuse the agent during execution.
- **Web agents often lack state assessment and replanning capabilities.** As illustrated in Figure 8, when the agent enters a product detail page to verify a 1-year warranty, it fails to reassess its state upon realizing that the requirement is unmet. Instead of returning to the search results page, the agent continues to scroll within the current page, inefficiently attempting to locate an alternative product.
- **Web agents are limited by action space.** As shown in Figure 9, the agent attempts to filter cameras within the $100–$300 price range but fails to do so due to its inability to interact with dynamic UI elements such as price sliders. Instead, it clicks the adjacent "Go" button without adjusting the slider values, resulting in ineffective filtering. This highlights a fundamental limitation of current agents: the constrained action space prevents them from performing fine-grained interactions required in realistic web environments.
- **Web agents lack the ability to learn from execution.** As illustrated in Figure 10, we present screenshots of the web agent across four different tasks. A consistent failure pattern emerges: the agent repeatedly uses keyword retrieval but fails to satisfy the filtering or sorting constraints in these tasks. The retrieved results do not meet the

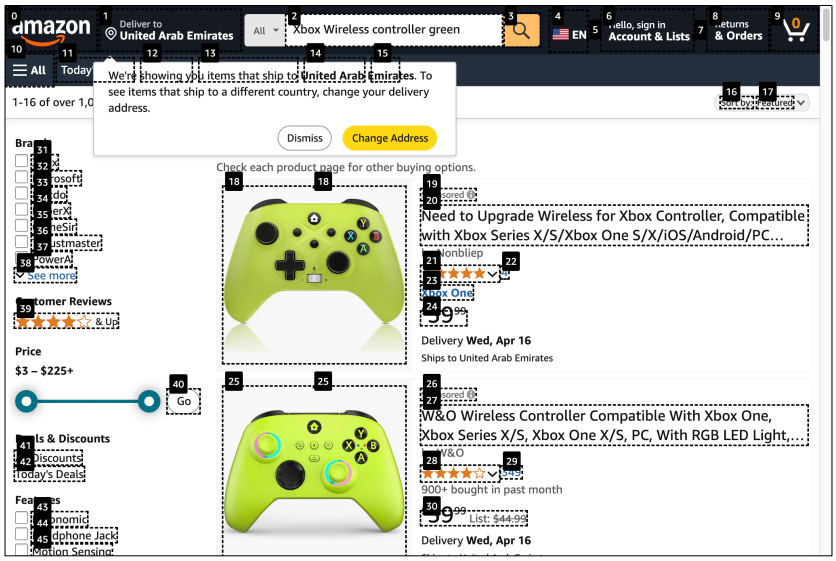


Fig. 7. Limited grounding ability of web agents.

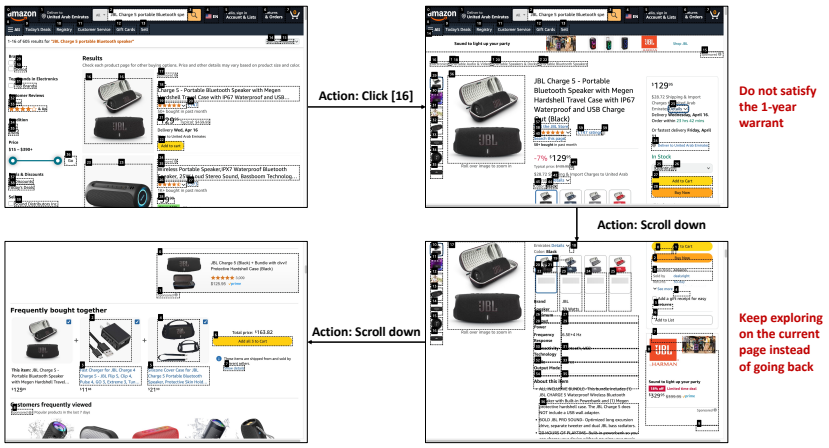


Fig. 8. Illustration of web agent's failure to reassess and replan.



Fig. 9. Illustration of the web agent's failure to apply the price filter during task execution.

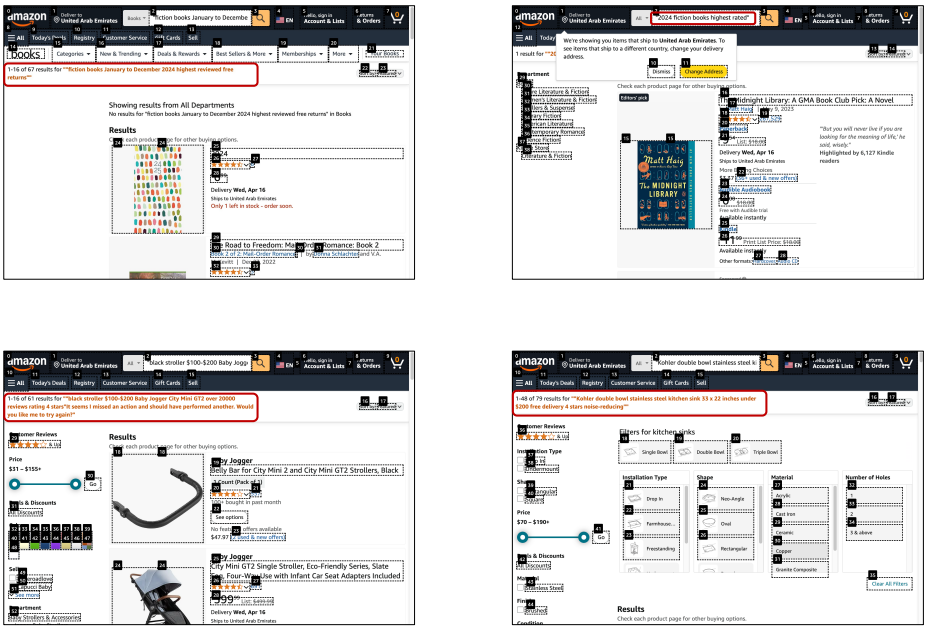


Fig. 10. Illustration of the web agent's failure to learn from execution.

corresponding requirements. This repetition across tasks highlights the agent's lack of execution-time learning, preventing it from abstracting past mistakes and adapting its behavior accordingly.

– **Earlier commercial-system cases illustrate hallucination errors.** As shown in Figure 11, the task requires *identifying a Women's Vintage Floral Maxi Dress in Navy Blue, Size: Medium, and explaining the return policy if free returns are available*. However, as shown in Figure 12, the first link returned by the OpenAI deep research system points to a product only available in Large and XX-Large sizes. Despite this mismatch, the system hallucinates that the size requirement is met. Moreover, Link2 and Link3 direct to non-Amazon websites, which violates the task constraint. In an attempt to explain the return policy, the system incorrectly extracts return information from these external sites. This case illustrates hallucinations in satisfying attribute constraints and sourcing policy information in the original commercial version.

Task:
Locate the Return Policy for Women's Vintage Floral Maxi Dress on Amazon. Color: Navy Blue, Size: Medium. If free returns are available, please explain the return process for this item.

OPENAI deep research system answer:
Understood. I will search Amazon for listings of a Women's Vintage Floral Maxi Dress in Navy Blue, size Medium, and identify the return policy for each listing. I will focus on whether free returns are available and explain the return process for the item. I'll share the direct Amazon links for all relevant listings I find.

Research completed in 11m:

Return Policy for Women's Vintage Floral Maxi Dress on Amazon
Free Return Policies for Navy Blue Vintage Floral Maxi Dresses (Medium)
When shopping on Amazon for a women's vintage floral maxi dress in Navy Blue (size Medium), it's important to check if the item offers Free Returns. Below are a few popular listings that match this description, along with their return policy details:
Floerns Women's Sleeveless Halter Neck Vintage Floral Print Maxi Dress (Navy Blue)
Product: *Floerns Women's Sleeveless Halter Neck Vintage Floral Print Maxi Dress*
Color: "A Navy Flower",
Size: Medium.
Link: Amazon Product Page – Floerns Halter Neck Floral Maxi
Free Returns: Yes. This Floerns dress is Prime-eligible and marked with "FREE Returns", meaning you can return it at no cost if it doesn't fit or meet your expectations [Link1]. Amazon provides at least one prepaid return shipping option for eligible items, so returning this dress is hassle-free.
Return Process: You have 30 days from delivery to initiate a return via your Amazon account. Simply go to Your Orders, select this dress, and choose "Return or Replace Items."
Because it's eligible for free returns, Amazon will offer a prepaid shipping label or QR code for drop-off [Link2]. You can then package the dress (in new, unworn condition with tags) and drop it off at an approved location (e.g. UPS Store, Whole Foods, Kohl's, etc.) without paying return shipping [Link3]. Once the item is received, you'll get a full refund to your original payment method.

[Link1]: https://www.amazon.com/Floerns-Womens-Sleeveless-Halter-Vintage/dp/B0C3HQLBBQ#:~:text=,least%20one%20free%20return
[Link2]: https://www.smoking-er.com/store/rosemary-pellet#:~:text=,preferred%20free%20shipping%20option%20Drop%C2%A0off%C2%A0and%C2%A0leave
[Link3]: https://www.smoking-er.com/store/rosemary-pellet#:~:text=,preferred%20free%20shipping%20option%20Drop%C2%A0off%C2%A0and%C2%A0leave

Fig. 11. Illustration of hallucination errors in the OpenAI deep research system. The system returns three links for the query, including two that point to non-Amazon websites, and incorrectly assumes the returned products match the specified size and platform constraints.

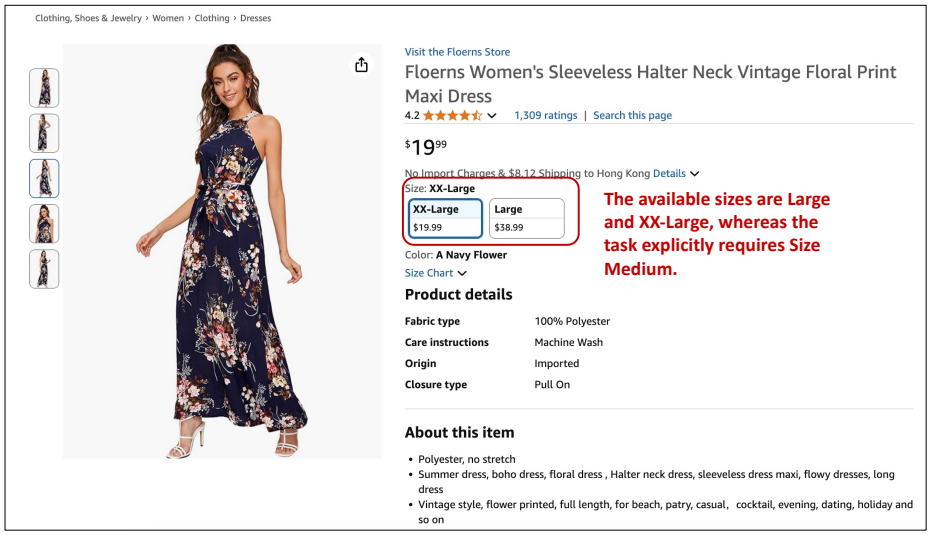


Fig. 12. Detailed view of the first returned product link. Although the task specifies size Medium, the linked product only offers Large and XX-Large options.

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